



Convolutional attention-based adaptive separation network for EEG artefact removal

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ABSTRACT

Electroencephalography (EEG) is an important electrical signal for recording physiological activities of the brain. Usually, its high temporal resolution can be susceptible to contamination by artefacts such as electrooculogram (EOG) and electromyogram (EMG), which affects the subsequent signal processing and analysis. Therefore, EEG artefact removal is particularly important and can lay the foundation for brain-computer interface (BCI) applications. However, most of the existing studies suffer from insufficient ability to extract global and local coarse-grained and fine-grained features interactively, insufficient effective constraints on latent features, and excessive computational complexity. To address these problems, this study designs a convolutional attention-based adaptive separation network (ASNet) for EEG artefact removal. Here, a separator is introduced based on the U-net architecture to constrain the potential features so as to adaptively separate the required discriminative features. In addition, this study improves on the disadvantages of the Transformer itself by designing a novel convolutional attention module to improve the coarse-grained and fine-grained feature interaction learning capability while reducing the number of parameters and computational effort. The experimental results show that ASNet achieves impressive artefact removal performance on fully synthetic datasets, semi-synthetic datasets, and real datasets, and has low computational complexity, which is a significant advantage over existing state-of-the-art methods. The codes are available at <https://github.com/qwertwjq/ASNet/tree/main>.

1. Introduction

Electroencephalography (EEG), which is the recording of electrical signals from the electrophysiological activity of the cerebral cortex by means of sensors on the scalp [1,2], is the main method of non-invasive acquisition of brain signals [3–5]. It is of great significance in the fields of cognitive psychology [6,7], brain-computer interface (BCI) [8,9] and diagnosis of neurological disorders [10,11]. However, EEG signals are weak and have high temporal resolution, and are easily interfered with by noise generated by experimental equipment and artefacts caused by physiological activities [12,13]. The sources of these artefacts include measurement instruments and human activities [1,14–16]. Among them, artefacts generated by measurement instruments can be avoided by accurate recording systems and strict recording procedures. However, the generation of physiological artefacts is complex and not easy to eliminate, and they include EOG [17], EMG [18], ECG [19],

and external artefacts (motion artefacts, vascular wave artefacts). It has been shown that physiological artefacts can seriously affect the acquisition of the true characteristics of EEG signals thereby interfering with the subsequent analysis of EEG signals [20,21].

The main challenge in EEG artefact removal is to identify and remove the artefacts while preserving the underlying neural activity information as much as possible. Traditional artefact removal methods mainly include regression, filtering [22], blind source separation (BSS) [23], wavelet transform (WT) [24], empirical modal decomposition [25], and hybrid methods [26]. However, most of these methods rely on prior analytical experience and are difficult to remove EEG artefacts automatically. So they are only applicable to specific types of artefact removal, and the problem of arranging and selecting the decomposed EEG components and artefact components is still not effectively solved. Moreover, traditional methods have high computational

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complexity and are not suitable for online applications [27]. To address these challenges, Gajbhiye et al. [28] developed the Multi-Resolution Total Variation (MTV) and Multi-Resolution Weighted Total Variation (MWTV) methods, which utilize Discrete Wavelet Transform (DWT) to effectively remove motion artefacts from single-channel EEG signals through multi-resolution analysis, filtering, and reconstruction techniques to enhance signal quality. On the other hand, Sharma et al. [29] proposed a framework combining Discrete Fourier Transform (DFT) with Adaptive Chirp Mode Decomposition (ACMD) for the automatic detection and removal of ocular artefacts (OAs) from single-channel EEG signals. This process includes frequency domain processing, signal decomposition, and mode selection based on a peak-count threshold criterion to purify the EEG signals. Despite these advancements improving artefact removal to some extent, EEG artefact removal methods still require better efficiency in terms of time and accuracy. The current research focus is on leveraging deep learning methods to achieve effective automatic artefact removal, aiming for higher precision and real-time processing capabilities.

With the rapid development of deep learning, it has been widely used in tasks such as image denoising and speech denoising in recent years [30,31]. Meanwhile, inspired by these tasks, it is also reflected in EEG artefact removal. Gao et al. [32] proposed a dual-scale model combining convolution and LSTM, which is conducive to spatio-temporal feature extraction of EEG signals to enhance model denoising effect. Sawangjai et al. [33] proposed a model based on the Generative Adversarial Net to remove ocular artefacts and adaptively generate pure EEG signals from noise-containing signals. Yi et al. [34] designed a denoising model based on transformer architecture, which slices the EEG signals as inputs and reduces the model complexity. Chen et al. [35] designed a variational self-scaling model combined with U-Net, Transformer, a variational self-coding model, which has a better denoising effect compared with the advanced methods in recent years. Zhang et al. [36] designed a multi-module denoising model, which passes the output of the previous module to the next module as an input, and achieves a good denoising effect. Yu et al. [37] proposed a combination of encoding, decoding and decomposition, which uses the features obtained from encoding to be multiplied with the decomposition of the obtained attenuation vectors are multiplied and then decoded, which can generate pure EEG signals and artefacts.

In addition, Transformer has been widely used in the fields of natural language processing and computer vision in recent years, and its attention mechanism is a key part in the design of the internal structure. The self-attention mechanism used by Transformer [38] is quadratic in computational complexity, and it brings problems such as high computational complexity, high memory usage, and inefficiency in predicting the output of long sequences. The Transformer model has advantage of capturing content-based global interactions, but its extraction of local features is insufficient. Therefore, there are many researches to optimize for its structure. Zhou et al. [39] designed an Informer structure to halve the output of each self-attention layer of the Transformer model to highlight the main attention and reduce the computational complexity of the self-attention mechanism. Yu et al. [40] designed a Metaformer model to validate that the model's multi-head self-attention module can be replaced with spatial MLP or PoolFormer simultaneously to achieve better results and lower computational complexity. Gulati et al. [41] designed a Conformer model, which integrates the Transformer's goodness at capturing global features and CNN's effective use of local features. The best results were achieved by combining convolutional neural network and Transformer to model the local and global dependencies of sequence information. Liu et al. [42] designed a Swin transformer model, which is based on the ViT model, and introduces a sliding window mechanism to allow the model to learn information across the window. At the same time, through the downsampling layer, the model can save the computation as well as focus on the local and global features. Chen et al. [43] improved the structure of the Transformer model by reducing the

computational complexity while ensuring the effectiveness, using a combined convolutional and channel attention module instead of a multi-head self-attention module. The gate control unit (GLU) is introduced to replace the RELU activation and simplified to improve the model effect.

It is worth noting that the U-Net architecture has been shown to be effective for denoising, and combining it with the Transformer can further improve the denoising effect of the model [44]. Cao et al. [45] designed a Swin-Unet model for image denoising. They replaced the convolution in the Unet architecture with the Swin-Transformer, and introduced the sliding window, relative position encoding and hierarchical structure. Liu L et al. [46] designed a speech denoising model MFNet and used STDCT to get real 2D data. They also replaced the ordinary convolution with global and local feed-forward module (GLFB) module and reduced the number of model parameters based on U-Net architecture, which has similar global and local modelling capabilities as transformer.

By way of overview, we found that most of these denoising methods use reconstructed EEG generation, with fewer methods for separation of latent features. The model structure mostly uses convolutional neural networks, with only a few of these methods combining convolution with sequence models such as LSTM and Transformer. Previous studies have shown that the combination of convolution and sequence models can extract and fuse local and global features well [31]. However, most of the existing studies suffer from insufficient ability to extract global and local coarse-grained and fine-grained features interactively, and the computational complexity of the models is too high.

In this paper, a convolutional attention-based adaptive separation network (ASNet) is proposed for EEG artefact removal. The network mainly consists of three main functional blocks, which are encoding module, separation module and decoding module. Encoding module: it contains an input module, three sub-modules combined by multiple convolutional attention modules, and three down-sampling modules that abstract the input contaminated EEG to latent features. Separation module: it contains a sub-module of a combination of multiple temporal convolution modules, and a module of a combination of multiple dilated convolutions with progressively wider spacing of convolution kernels. The weights of the magnitude of their correlation with the EEG signal features are obtained from the input latent features, and then multiplied onto the latent features to achieve adaptive separation of the EEG discriminative features. This module has a progressively expanding receptive field that allows the convolution module to capture global features. Decoding module: it contains an output module, three sub-modules combining multiple convolutional attention modules, and three up-sampling modules to decode the latent features into reconstructed EEG. At the same time, we also improve the disadvantages of Transformer itself by designing a novel convolutional attention module to improve the ability of coarse-grained and fine-grained feature interaction learning while reducing the number of parameters and computation. And it is used as the main part of the encoder and decoder in the U-net architecture, so that the model can capture the local and global temporal dependence of EEG information.

The main contributions of this study are as follows:

1. We design a separator consisting of a combination of multiple expansive convolutions with progressively wider spacing of convolution kernels with progressively wider receptive fields, allowing the convolution module to capture global features.
2. We introduce the separator in the U-net architecture to constrain the latent features, and then better separate the features of pure signals from the noise features, and to adaptively capture the required discriminative features.
3. We compensate for the shortcomings of the Transformer by designing a novel convolutional attention module to improve coarse-grained and fine-grained feature interaction learning capabilities while reducing the number of parameters and computational effort. In addition, we incorporate it into the U-net

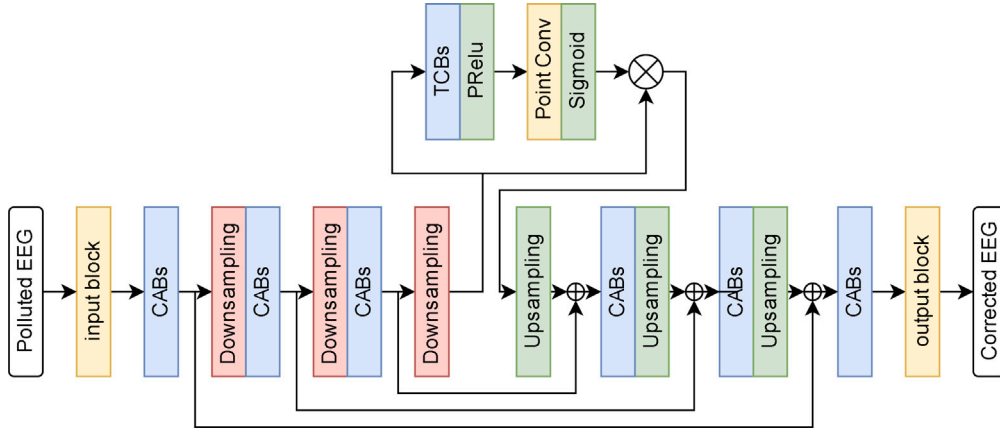


Fig. 1. ASNet is a network structure designed for removing EEG artefacts, consisting of three main modules: encoding, separation, and decoding. The encoding module converts contaminated EEG signals into latent features using an input module and multiple convolutional attention submodules (CABS). The separation module adaptively separates the latent features through temporal convolutions and pointwise convolutions. Finally, the decoding module reconstructs the separated features into pure EEG signals via upsampling and convolutional attention submodules.

architecture to better capture the local and global temporal dependency of EEG information.

4. The proposed ASNet achieves impressive artefact removal performance on fully synthetic datasets, semi-synthetic datasets, and real datasets, and has low computational complexity, which is a significant advantage over existing state-of-the-art methods.

2. Methods

Here, this study proposes a convolutional attention-based adaptive separated U-network for EEG signal artefact removal, called ASNet. ASNet is an end-to-end network that reconstructs pure EEG signals from original contaminated EEG signals by learning nonlinear transformations. In the next subsections, we describe ASNet in detail in terms of model structure, training and testing, experimental dataset and validation metrics.

2.1. Model structure

This section describes the structure of ASNet. As shown in Fig. 1, ASNet has three main functional blocks, encoding module, separation module and decoding module. The input is a single channel of contaminated EEG signal and the output is a pure EEG signal. The encoding module consists of an input module, three sub-modules (CABS) combining multiple convolutional attention modules, and three downsampling modules that encode the input contaminated EEG into latent features. The separation module includes a sub-module combining multiple temporal convolution modules, point convolution, etc., which obtains weights from the input latent features in terms of the magnitude of their correlation with the EEG signal features, and then multiplies them onto the latent features to achieve adaptive separation of the latent features. The decoding module contains one output module, three sub-modules (CABS, containing multiple convolutional attention modules) and three upsampling modules to decode the latent features into reconstructed EEG.

2.1.1. Encoding modules

2.1.1.1. Input module. This module constructs a standard one-dimensional convolution with a convolutional kernel size of 3 to convert one-dimensional data into two-dimensional data with two dimensions, sequence and channel, as shown in the following equation:

$$X = \text{InputBlock}(PE) \quad (1)$$

where PE denotes the polluted EEG signals, X denotes the input matrix, and the shape of the data variables: $PE \in R^{1 \times S}$ (S denotes the sequence length), $X \in R^{C \times S}$ (C denotes the channel dimension). *InputBlock* denotes the input module. For the input one-dimensional EEG data with shape R^S , the dimension is extended to obtain contaminated EEG with shape $R^{1 \times S}$. The input is fed to the *InputBlock* to get X .

2.1.1.2. Submodules and downsampling modules. The sub-module contains several convolutional attention modules, and the down-sampling module uses standard one-dimensional convolution to halve the sequence length and double the channel dimension. The encoding process for the three submodules (CABS) and the three downsampling modules is shown in the following equation:

$$E1 = \text{CABS}(X) \quad (2)$$

$$E2 = \text{CABS}(\text{DownSampling}(E1)) \quad (3)$$

$$E3 = \text{CABS}(\text{DownSampling}(E2)) \quad (4)$$

$$M1 = \text{DownSampling}(E3) \quad (5)$$

where X denotes the 2D input matrix, $E1$, $E2$, $E3$ denote the coded features in the encoding process, $M1$ denotes the latent features, and the shapes of the data variables: $X \in R^{C \times S}$, $E1 \in R^{C \times S}$, $E2 \in R^{2C \times \frac{S}{2}}$, $E3 \in R^{4C \times \frac{S}{4}}$, $M1 \in R^{8C \times \frac{S}{8}}$. *CABS* denotes the sub-modules, and *DownSampling* denotes the down-sampling modules. As shown in Eq. (2), the feature matrix X is first passed into the submodule to get $E1$; as shown in Eq. (3), $E1$ is input to the downsampling module and then passed into the submodule to encode to get $E2$; as shown in Eq. (4), $E2$ is input to the downsampling module and then passed into the submodule to get $E3$; as shown in Eq. (5), $E3$ is passed into the downsampling module to get the latent feature $M1$.

Among them, the design of the CABS is based on the Transformer architecture that includes the multi-headed self-attention module and the feedforward network module. The self-attention mechanism used by the multi-headed self-attention Module is of quadratic computational complexity, making it unsuitable for mobile devices or resource-constrained devices, and then results insufficient extraction of local features. To solve this problem, as shown in Fig. 2(a), the module constructs a combination of depth-separable convolution [47] with attention to replace the multi-head attention module, containing layer specification, point-by-point convolution, depth convolution, gate activation function, channel attention module, spatial attention module. Replace the

feedforward layer by constructing two point-by-point convolutions containing layer specification, a point-by-point convolution, gate activation function, and a point-by-point convolution. The aim of the construction is to ensure effective extraction of global and local features while reducing the number of parameters and computation. The calculated process of CABS is as follows:

$$F2 = DWConv(PointConv(LN(F1))) \quad (6)$$

$$F3 = GA(F2) \quad (7)$$

$$F4 = SA(CA(F3)) \quad (8)$$

$$F5 = PointConv(F4) + F1 \quad (9)$$

$$F6 = PointConv(LN(F5)) \quad (10)$$

$$F7 = GA(F6) \quad (11)$$

$$F8 = PointConv(F7) + F5 \quad (12)$$

where $F1$ denotes the input feature of convolutional attention module, $F2, F3, F4, F5, F6, F7$ denote the intermediate features, $F8$ denotes the output feature, and the shape of the data variable: if the input feature $F1$ is shaped as $R^{C \times S}$, then $F2 \in R^{2C \times S}$, $F3 \in R^{C \times S}$, $F4 \in R^{C \times S}$, $F5 \in R^{C \times S}$, $F6 \in R^{2C \times S}$, $F7 \in R^{C \times S}$, $F8 \in R^{C \times S}$. LN denotes layer normalization, $PointConv$ denotes point-by-point convolution, and $DWConv$ denotes deep convolution. GA denotes gate activation, CA denotes channel attention module, and SA denotes spatial attention module. As shown in Eq. (6), the input feature $F1$ is firstly layer normalized, and then it undergoes point-by-point convolution to double the channel dimension, and then it undergoes deep convolution to obtain feature $F2$. As shown in Eq. (7), feature $F2$ undergoes gate activation, and the channel dimension is halved to obtain feature $F3$. As shown in Eq. (8), the feature $F3$ undergoes the extraction of features by the channel attention module and the spatial attention module successively, and then the feature $F3$ obtains feature $F4$. As shown in Eq. (9), after point-by-point convolution of feature $F4$, plus input feature $F1$ to get feature $F5$, to achieve the residual mechanism. As shown in Eq. (10), layer normalization is performed on feature $F5$, and then point-by-point convolution is used to double the channel to obtain feature $F6$. As shown in Eq. (11), feature $F6$ is activated by a gate, and the dimension of the channel is halved to obtain feature $F7$. As shown in Eq. (12), point-by-point convolution is performed on feature $F7$, and then feature $F5$ is added to obtain feature $F8$, which implements the residual mechanism.

Depth separable convolution consists of point-by-point convolution and depth convolution. Point-by-point convolution is used to extract channel features, and all channels are aggregated, which can increase or decrease the feature dimension. Depth convolution is used to extract spatial features, grouping convolutions in feature dimensions and performing independent depth-by-depth convolution for each channel. Compared to standard convolution, it has the same feature extraction effect, but with low computational effort and low parameter count.

As shown in Fig. 2(b), the channel attention module designed in this study consists of global average pooling (GAP), pointwise convolution and Sigmoid activation function. The global average pooling is performed on the feature map to obtain one-dimensional channel information, and the channel information is multiplied by the dot convolution and Sigmoid activation function to obtain the channel attention weights, and then multiplied onto the channel to learn the channel relationship between the convolutional features. The channel attention module formula is as follows:

$$FC2 = GAP(FC1) \quad (13)$$

$$FC3 = FC1 * Sigmoid(Conv(FC2)) \quad (14)$$

where $FC1$ denotes input feature, $FC3$ denotes output feature. $FC2$ denotes intermediate feature. GAP denotes global average pooling. $Sigmoid$ denotes sigmoid activation function, and $Conv$ denotes convolution. Shape of data variables: if the input feature $FC1$ is shaped as $R^{C \times S}$, then $FC2 \in R^{C \times 1}$, $FC3 \in R^{C \times S}$. As shown in Eq. (13), the input feature $FC1$ is globally averaged pooled to obtain the feature $FC2$. Convolution and activation are performed successively on $FC2$ without changing its shape as shown in Eq. (14). The channel attention weights are obtained, and then the weights are multiplied to each channel of $FC1$ to obtain feature $FC3$.

The spatial attention module is constructed to learn the spatial relationship between convolutional features, as shown in Fig. 2(c). Global maximum pooling (GMP) and global average pooling (GAP) are performed on the channel information, so that features with more useful information are concentrated. Meanwhile, features without useful information are filtered, resulting in two feature maps representing different information. Then they are merged and then fused by pointwise convolution, and finally overlaid on the original input feature map through the Sigmoid operation to generate a weight map, thus making the target region enhanced. The spatial attention module formula is as follows:

$$FSA = GAP(FS1) \quad (15)$$

$$FSM = GMP(FS1) \quad (16)$$

$$FS2 = Concat(FSA, FSM) \quad (17)$$

$$FS3 = FS1 * Sigmoid(PointConv(FS2)) \quad (18)$$

where $FS1$ denotes input features, $FSA, FSM, FS2$ denotes intermediate features, and $FS3$ denotes output features. GAP denotes Global Average Pooling, GMP denotes Global Maximum Pooling, and $Concat$ denotes the concatenation of sequence dimensions. $PointConv$ denotes pointwise convolution, and $Sigmoid$ denotes the Sigmoid activation function. Shape of data variables: if the input feature $FS1$ is shaped as $R^{C \times S}$, then $FSA \in R^{1 \times S}$, $FSM \in R^{1 \times S}$, $FS2 \in R^{2 \times S}$, $FS3 \in R^{C \times S}$. As shown in Eq. (15), global average pooling is performed on the channel information of the input feature $FS1$ to obtain the feature FSA . As shown in Eq. (16), global maximum pooling to get feature FSM . As shown in Eq. (17), join feature FSA, FSM in channel dimension to get feature $FS2$. As shown in Eq. (18), perform dot convolution on feature $FS2$ to halve the channel dimension, activate to get the weight of the sequence region, and multiply to the sequence dimension of input feature $FS1$ to get $FS3$.

In addition, Gated Linear Unit (GLU) [48] is introduced to improve the expressiveness and performance of the model. GLU halves the number of channels, splits the feature into two parts with the same number of channels in the channel dimension, and multiplies them element by element. It is possible to efficiently learn which information should be retained or suppressed and reduce the amount of computation. The gate activation function formula is as follows:

$$FG = FG1 \odot FG2 \quad (19)$$

where $FG1, FG2$ denote the input features are divided into two parts of features with the same number of channels in the channel dimension. $\odot$ denotes the Hadamard product and FG denotes the output feature. Shape of data variables: if the shape of input features is $R^{C \times S}$, then $FG1 \in R^{\frac{C}{2} \times S}$, $FG2 \in R^{\frac{C}{2} \times S}$, $FG \in R^{\frac{C}{2} \times S}$. As shown in Eq. (19), the features $FG1$ and $FG2$ are subjected to the Hadamard product to obtain the feature FG .

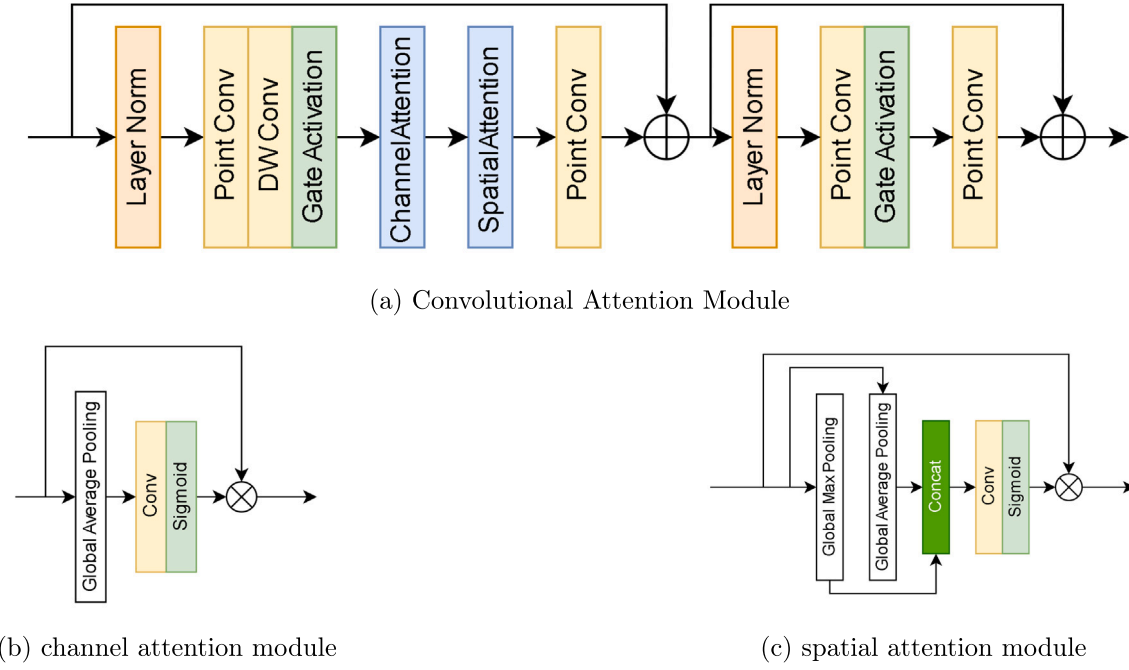


Fig. 2. Design of the Convolutional Attention Submodule (CABS). It shows the design details of CABS. (a) Describes the overall structure of CABS, which combines depthwise separable convolutions with attention mechanisms to replace the multi-head self-attention module and uses pointwise convolutions to substitute the feedforward network layer; (b) Details the channel attention module, which generates channel weights using global average pooling (GAP), pointwise convolution, and Sigmoid activation to learn inter-channel relationships; (c) Describes the spatial attention module, which processes channel information through global max pooling (GMP) and global average pooling (GAP) to generate two feature maps, fuses them via pointwise convolution, and overlays the resulting spatial weight map onto the original feature maps to enhance target region features.

2.1.2. Separation module

This module includes PRelu activation, point convolution, Sigmoid activation function, and several temporal convolution modules. We input the latent features these modules to get the weights of the features related to the EEG signals, and then multiply them to the latent features to achieve adaptive separation of the latent features. The separation process is shown in the following equations:

$$F = PRelu(TCBS(M1)) \quad (20)$$

$$W = Sigmoid(PointConv(F)) \quad (21)$$

$$M2 = W \odot M1 \quad (22)$$

where $M1$, $M2$ denote the latent features, F denotes the features extracted by the sub-modules, $\odot$ denotes the Hadamard product, W denotes the weights of the features related to the EEG signals, and the shapes of the data variables are $M1 \in R^{8C \times \frac{S}{8}}$, $M2 \in R^{8C \times \frac{S}{8}}$, $F \in R^{8C \times \frac{S}{8}}$, $W \in R^{8C \times \frac{S}{8}}$. $TCBS$ denotes the sub-modules and $PointConv$ denotes point convolution. As shown in Eq. (20), the latent feature $M1$ is extracted by the sub-module and activated by PRelu to obtain F ; as shown in Eq. (21), the weight W is obtained by point convolution of the feature F and activated by Sigmoid; as shown in Eq. (22), Finally the weight matrix W and the latent feature matrix $M1$'s are subjected to Hadamard product to obtain $M2$.

As shown in Fig. 3(a), the temporal convolution module consists of a combination of multiple dilated convolution modules (DCBs) with gradually expanding convolution kernel spacing. The convolution kernel spacing increases exponentially, the value of d is used in the figure to represent the convolution kernel spacing, the model is designed to have a sequence dimension of 64 for the latent features, a one-dimensional convolution kernel size of 3, and a maximum convolution kernel spacing of 2^i , i in the model is set to be 4 that is, the maximum convolution kernel spacing is 16. As shown in Fig. 3(b), the dilated convolution block consists of depth separable convolution and dot convolution. The input is added to the output using residual connections. It contains

point-by-point convolution, layer specification, PRelu function, depth extended convolution, and point-by-point convolution. The extended convolution module formula is as follows:

$$TC2 = LN(PointConv(TC1)) \quad (23)$$

$$TC3 = DWConv_i(PRelu(TC2)) \quad (24)$$

$$TC4 = PointConv(TC3) + TC1 \quad (25)$$

where $TC1$ denotes input features, $TC2$ and $TC3$ denote intermediate features, $TC4$ denotes output features, LN denotes layer normalization, and $DWConv_i$ denotes dilated convolution with convolution kernel spacing of 2^i . Shape of data variables: if the input feature $TC1$ is shaped as $R^{C \times S}$, then $TC2 \in R^{2C \times S}$, $TC3 \in R^{2C \times S}$, $TC4 \in R^{C \times S}$. As shown in Eq. (23), firstly, point convolution is carried out on the input feature $TC1$ and then layer normalization is carried out to obtain the feature $TC2$. As shown in Eq. (24), PRelu activation is carried out on feature $TC2$ and then dilatational convolution is carried out with a convolution kernel spacing of 2^i . As shown in Eq. (25), the output feature $TC4$ is obtained by adding the input feature $TC1$ after pointwise convolution of feature $TC3$.

2.1.3. Decoding module

2.1.3.1. Submodules and upsampling modules. The sub-module contains several convolutional attention modules, and the upsampling module uses inverse convolution to double the sequence dimension and halve the channel dimension. The decoding process for the three submodules (CABS) and the three upsampling modules is shown in the following equations:

$$M3 = UpSampling(M2) \quad (26)$$

$$D1 = UpSampling(CABS(M3 + E3)) \quad (27)$$

$$D2 = UpSampling(CABS(D1 + E2)) \quad (28)$$



Fig. 3. Design of the temporal convolution module. It shows the design of the temporal convolution module. (a) Displays the structure composed of multiple dilated convolution blocks (DCBs) with progressively increasing kernel spacing to capture long-range dependencies. (b) Details the structure of a single dilated convolution block, including depthwise separable convolution, pointwise convolution, layer normalization, PReLU activation functions, and residual connections, aiming to enhance feature representation while maintaining low computational complexity.

$$D3 = CABS(D2 + E1) \quad (29)$$

where $M2$, $M3$ denote the latent features and $D1$, $D2$, $D3$ denote the decoded features in the decoding process. The shapes of data variables are $M2 \in R^{8C \times \frac{S}{8}}$, $E3 \in R^{4C \times \frac{S}{4}}$, $D1 \in R^{2C \times \frac{S}{2}}$, $D2 \in R^{C \times S}$, $D3 \in R^{C \times S}$. $CABS$ denotes the sub-module, and $UpSampling$ denotes the up-sampling module. As shown in Eq. (26), the latent feature $M2$ is first passed to the up-sampling module for processing to obtain $M3$; as shown in Eq. (27), $M3$ and $E3$ are summed and then input to the sub-module and passed to the up-sampling module to obtain $D1$; as shown in Eq. (28), $D1$ and $E2$ are summed and then input to the sub-module and then passed to the up-sampling module to obtain $D2$; as shown in Eq. (29), $D2$ and $E1$ are summed and then input to the sub-module to obtain $D3$.

2.1.3.2. Output module. This module constructs a standard one-dimensional convolution with a convolution kernel size of 3 to convert two-dimensional data with two dimensions, sequence and channel, into one-dimensional reconstructed EEG data, as shown in the following equation:

$$CE = OutBlock(D3) \quad (30)$$

where CE denotes the reconstructed EEG (Corrected EEG), $D3$ denotes the decoding features during the decoding process, and the shape of the data variables: $D3 \in R^{C \times S}$, $CE \in R^{1 \times S}$, and $OutBlock$ denotes the Output Module. $D3$ is passed into the Output Module to get CE , and the reconstructed EEG with shape $R^{1 \times S}$ is reduced in dimensions to get the reconstructed EEG with shape R^S .

2.2. Training and testing

The contaminated EEG signals are used as input and the reconstructed EEG signals are output to train the model in a supervised manner. The mean square error (MSE) between the output and the expected reconstructed pure EEG is used as the loss function. To train the network, the Adam optimization algorithm is used and the network hyperparameters are set as shown in Table 1. The EEG data and artefacts from the semi-synthetic dataset are mixed and then divided into training, validation, and test sets in a ratio of 8:1:1. The model is tested after training and can be used to remove ocular artefacts and EMG artefacts from single channel EEG.

2.3. Datasets

2.3.1. Fully synthetic dataset

According to Zangeneh et al. [49]'s description, we generated 19 segments of 10-s EEG signals, EMG artefact signals, and ECG artefact

Algorithm 1 Optimization Process of ASNet

Require: Clean EEG segments S and their corresponding contaminated EEG segment $\hat{S}$.

Ensure: Trained model parameters for ASNet.

for epoch do

for batch $B, \hat{B}$ in $S, \hat{S}$ do

Encode the input contaminated EEG signal $\hat{B}$ into features $E3$ using the encoding module (including Eq. (1),(2),(3),(4),(5)).

Input the features $E3$ into the separation module (including Eq. (20),(21),(22)) to obtain the adaptively separated artefact features as features $M2$.

Decode the features $M2$ into corrected EEG signals CE using the decoding module (including Eq. (9),(10),(11),(12),(13)).

Update the model parameters based on the Mean Square Error (MSE) between the Clean EEG segments B and the corrected EEG signals CE through AdamW optimizer.

end for

end for

Table 1

Hyperparameter settings for the network.

Hyper-parameter	Setting value
Number of CAB contained in submodule (CABS)	8
Number of TCB contained in the submodule (TCBS)	8
Batchsize	50
Length of the input segment	512
Size of the convolution kernel	3
Initial learning rate	0.0005
Epoch	200

signals, all sampled at 256 Hz. The pure EEG signal was modelled by superimposing four sine functions with random frequencies in the range of 4–30 Hz. EMG artefacts were simulated by band-pass filtering random noise in the 20–60 Hz range, while ECG artefacts were simulated by band-pass filtering random noise in the 1–3 Hz range, as shown in Fig. 4(a). By randomly mixing the pure EEG signal with both types of artefacts, we obtained fully synthetic contaminated EEG signals, as illustrated in Fig. 4(b).

2.3.2. Semi-synthetic dataset

A semi-synthetic dataset for EEG artefact removal was designed in EEGdenoiseNet [50], which contains 4514 pure EEG segments, 3400 eye artefact segments and 5598 muscle artefact segments. There are 512 samples (2 s) for each EEG segment, and the pure EEG segments are linearly mixed with EOG or EMG artefact segments to generate contaminated EEG signals with different signal-to-noise ratios (Eq. (32)) according to Eq. (31). The signal-to-noise ratio ranges from -7 to 2 dB. The semi-synthetic dataset of contaminated EEG obtained by mixing

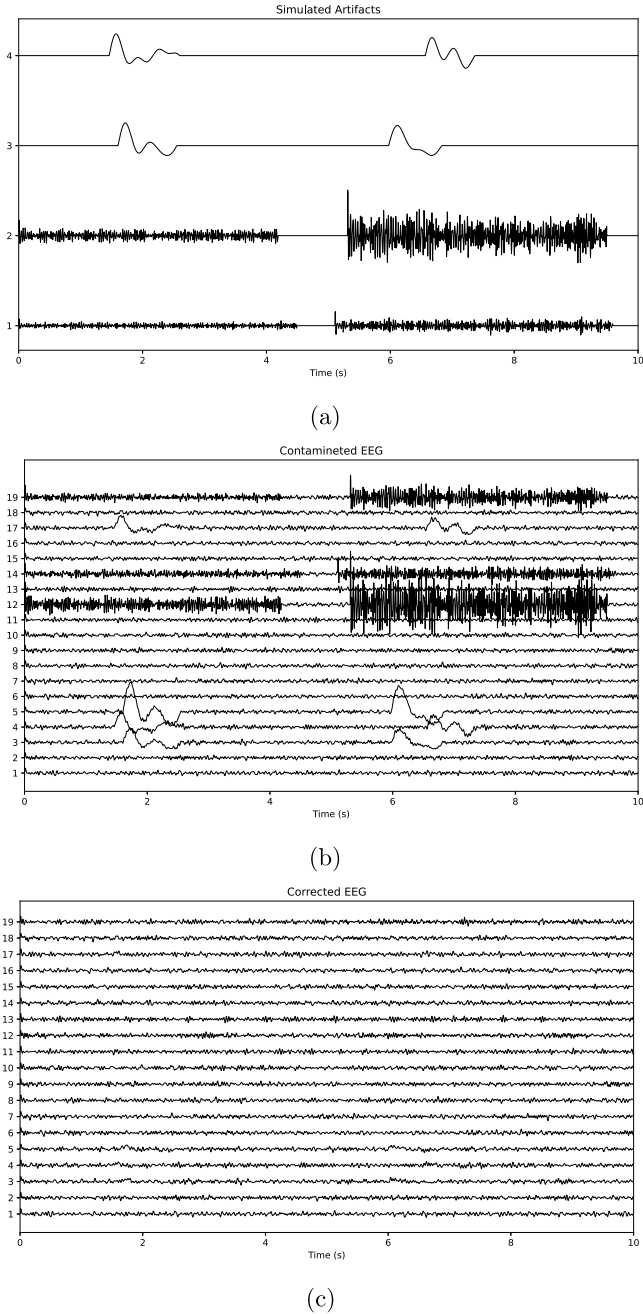


Fig. 4. Generation of fully synthetic datasets and denoising results of ASNet. It illustrates the generation process of fully synthetic datasets and the denoising results of ASNet. (a) Demonstrated the generation method of synthetic EMG and EOG artefacts; (b) Shows the contaminated EEG signals generated by randomly mixing these signals; (c) Presents the denoising results of ASNet, where EOG and EMG artefacts are effectively suppressed in the red-boxed regions, validating the model's effectiveness.

pure EEG and artefact is used to train the model.

$$y = x + \lambda \cdot n \quad (31)$$

where x represents the pure electroencephalogram (EEG) signal, which corresponds to the ideal state of EEG signals without any artefact interference; n represents artefacts (such as electrooculography or electromyography), which typically degrade the quality of EEG signals; and y represents the contaminated EEG signal obtained by mixing the pure EEG signal with artefacts at a specific weight. Here, λ is a proportionality coefficient used to adjust the intensity of the artefact

signal, determining the extent to which artefacts affect the pure EEG signal.

$$\text{SNR} = 10 \times \log \frac{\text{RMS}(x)}{\text{RMS}(\lambda \cdot n)} \quad (32)$$

where $\text{RMS}(x)$ denotes the root mean square (RMS) value of the pure EEG signal, which measures the average power level of the signal. It is calculated as the square root of the average of the squared values of all sampling points. $\lambda \cdot n$ is the result of multiplying the artefact signal by the noise intensity coefficient, reflecting the actual strength of the artefact signal; $\text{RMS}(\lambda \cdot n)$ represents the RMS value of the artefact signal, which measures its average power level. The final calculated signal-to-noise ratio (SNR) is expressed in decibels (dB) and quantifies the strength of the pure signal relative to background noise. A higher SNR value indicates better signal quality and lower noise impact, while a lower SNR value suggests that the signal contains more noise.

2.3.3. Real dataset

BCI Competition 2008 Graz Dataset A [51] covers EEG data from nine participants in an instruction-based BCI study involving four motor imagery tasks: left-handed, right-handed, bipedal and tongue. Each participant was tested twice over two days, each containing six rounds of 48 attempts each, totalling 288. Five minutes of EEG data were collected prior to testing to assess EOG effects, including eye opening, eye closing and eye movement sessions. Data were recorded through 22 10–20 system electrodes with a sampling rate of 250 Hz and band-pass filtering 0.5–100 Hz, equipped with a 50 Hz trap filter. Three EOG channels were also recorded to assist in artefact removal. The data were checked by experts to exclude bad trials and stored in GDF format with accompanying detailed labelling for BCI performance evaluation.

2.4. Evaluation metrics

In this study, four quantitative performance metrics are used to evaluate the model: relative root mean square error ($RRMSE$), correlation coefficient (CC), spectral relative root mean square error ($RRMSE_{\text{PSD}}$), and mutual information (MI).

1. $RRMSE$: In the scenario of removing EEG signal artefacts, $RRMSE$ measures the deviation between pure and contaminated signals and is calculated as follows:

$$\text{RRMSE}(s, \hat{s}) = \frac{\text{RMS}(s - \hat{s})}{\text{RMS}(s)} \quad (33)$$

where s denotes the pure EEG signals and $\hat{s}$ is the corrected EEG signals corresponding to s . RMS denotes the root mean square plant, which is the square root of the average value after summing the squares of each value. The performance of the artefact removal method increases when the $RRMSE$ value decreases.

2. CC : CC measures the similarity between pure and contaminated EEG segments and indicates the extent to which the corrected EEG segments contain valid information from the pure EEG segments. The calculation of CC can be defined as:

$$\text{CC}(s, \hat{s}) = \frac{\text{Cov}(s, \hat{s})}{\sqrt{\text{Var}(s) \text{Var}(\hat{s})}} \quad (34)$$

where Cov and Var denote the calculation of covariance and variance, respectively. It is worth noting that a higher CC means that the corrected EEG segments contain richer information.

3. $RRMSE_{\text{PSD}}$: $RRMSE_{\text{PSD}}$ is a metric for evaluating frequency-domain deviation, corresponding to the $RRMSE$ error in the time domain. It assesses artefact removal performance by calculating the difference in power spectral density (PSD) between the pure EEG signal and the reconstructed signal, with the formula:

$$\text{RRMSE}_{\text{PSD}} = \frac{\text{RMS}(\text{PSD}(\hat{s}) - \text{PSD}(s))}{\text{RMS}(\text{PSD}(s))} \quad (35)$$

Here, s represents the pure signal, $\hat{s}$ represents the reconstructed signal, and RMS denotes the root mean square value. A lower $\text{RRMSE}_{\text{PSD}}$ value indicates better artefact removal performance.

4. MI : Mutual Information (MI) measures the correlation between two signals, with the formula:

$$MI = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} p(\hat{s}, s) \log \left(\frac{p(\hat{s}, s)}{p(\hat{s})p(s)} \right) d\hat{s} ds \quad (36)$$

Here, $p(\hat{s}, s)$ is the joint probability density function, while $p(s)$ and $p(\hat{s})$ are the marginal probability density functions. s denotes the pure EEG signals and $\hat{s}$ is the corrected EEG signals corresponding to s . A higher MI value indicates greater similarity between the reconstructed signal and the pure signal, reflecting better method performance.

3. Experiments and results

3.1. Results on fully synthetic dataset

After applying the ASNet model to denoise the contaminated EEG signals, we obtained the corrected EEG signals, as shown in Fig. 4(c). In the red-boxed regions of the contaminated EEG signals, eye movement artefacts (EOG) and muscle artefacts (EMG) were effectively suppressed, demonstrating the significant ability of the ASNet model to remove ECG and EMG artefacts from fully synthetic datasets. Furthermore, the amplitudes of the EOG and EMG artefacts differed, further verifying the model's adaptability to different types of artefacts. The ASNet model is capable of removing synthetic artefacts of varying intensities while preserving the true EEG signals well.

3.2. Results on semi-synthetic EEG data

Here, we test the denoising performance of the model using the semi-synthetic EEG dataset, comparing the results with EEGDNet, Denoiseformer, DeepSeparator, ACMD, MTV and MWTv.

3.2.1. EOG artefact removal

The experiment quantitatively evaluated the performance of EOG artefact removal on a semi-synthetic dataset using four metrics: RRMSE, CC, $\text{RRMSE}_{\text{PSD}}$, and MI . As shown in Table 2, ASNet performed best across all metrics, achieving the lowest RRMSE and $\text{RRMSE}_{\text{PSD}}$, as well as the highest CC and MI . This indicates that ASNet outperforms other methods in EOG artefact removal. From the boxplots in Fig. 5, the distribution of ASNet's performance in terms of RRMSE, CC, $\text{RRMSE}_{\text{PSD}}$, and MI is more concentrated compared to other models, demonstrating its stronger robustness in EOG artefact removal. To further investigate the denoising robustness of the model, the experiment was conducted at multiple signal-to-noise ratio (SNR) levels, ranging from -7 dB to 2 dB. The quantitative results are shown in Fig. 6. Overall, the performance of all methods deteriorates as the SNR decreases. However, ASNet consistently outperforms other methods across different SNR levels, exhibiting lower RRMSE and $\text{RRMSE}_{\text{PSD}}$, as well as higher CC and MI . This indicates that ASNet can maintain robust performance even when the input signals contain significant EOG artefacts.

As shown in Fig. 7, the best and worst results of removing EOG artefacts using the ASNet model are presented. The figure displays time-domain and frequency-domain segments, particularly highlighting the performance in the 0–10 Hz band. The power spectral density (PSD) of EEG signals contaminated by EOG artefacts is significantly higher than that of clean EEG signals. After denoising, the PSD in the 0–10 Hz band is substantially reduced. Within the 0–80 Hz band, the PSD of the denoised EEG signals closely matches that of the clean EEG signals, demonstrating the ASNet model's effectiveness in retaining true EEG signals during the denoising process. Since EOG artefacts primarily belong to low-frequency noise, the ASNet model effectively suppresses artefacts in the low-frequency band while preserving the key features

Table 2

Comparison of eog removal results. The mean $\pm$ std of four metrics from test samples are presented.

Metric	RRMSE	$\text{RRMSE}_{\text{PSD}}$	CC	MI
EEGDNet	0.384 $\pm$ 0.031	0.446 $\pm$ 0.032	0.912 $\pm$ 0.024	5.23 $\pm$ 0.748
Denoiseformer	0.292 $\pm$ 0.014	0.362 $\pm$ 0.013	0.951 $\pm$ 0.012	6.21 $\pm$ 0.336
DeepSeparator	0.612 $\pm$ 0.063	0.652 $\pm$ 0.062	0.769 $\pm$ 0.091	4.93 $\pm$ 1.047
ACMD	0.273 $\pm$ 0.026	0.241 $\pm$ 0.025	0.945 $\pm$ 0.016	6.38 $\pm$ 0.328
MTV	0.295 $\pm$ 0.024	0.258 $\pm$ 0.023	0.942 $\pm$ 0.019	6.23 $\pm$ 0.275
MWTv	0.293 $\pm$ 0.022	0.259 $\pm$ 0.022	0.943 $\pm$ 0.021	6.24 $\pm$ 0.234
ASNet	0.252 $\pm$ 0.018	0.213 $\pm$ 0.017	0.966 $\pm$ 0.011	6.43 $\pm$ 0.336

Table 3

Comparison of emg removal results. The mean $\pm$ std of four metrics from test samples are presented.

Metric	RRMSE	$\text{RRMSE}_{\text{PSD}}$	CC	MI
EEGDNet	0.624 $\pm$ 0.068	0.626 $\pm$ 0.068	0.691 $\pm$ 0.236	4.292 $\pm$ 0.628
Denoiseformer	0.372 $\pm$ 0.022	0.432 $\pm$ 0.024	0.894 $\pm$ 0.064	5.213 $\pm$ 0.216
DeepSeparator	0.712 $\pm$ 0.083	0.762 $\pm$ 0.083	0.634 $\pm$ 0.231	3.733 $\pm$ 1.024
ACMD	0.333 $\pm$ 0.016	0.384 $\pm$ 0.015	0.904 $\pm$ 0.015	5.445 $\pm$ 0.323
MTV	0.331 $\pm$ 0.021	0.399 $\pm$ 0.024	0.903 $\pm$ 0.013	5.338 $\pm$ 0.332
MWTv	0.339 $\pm$ 0.018	0.391 $\pm$ 0.019	0.897 $\pm$ 0.009	5.341 $\pm$ 0.337
ASNet	0.328 $\pm$ 0.019	0.373 $\pm$ 0.018	0.915 $\pm$ 0.016	5.532 $\pm$ 0.316

of the true signal. However, potential failure cases remain unavoidable. As shown in the figure, in the worst-case scenario, there is a significant discrepancy between the waveforms of the denoised EEG signals and the clean EEG signals in the time domain; in the frequency domain, the PSD of the denoised EEG signals in the 0–80 Hz band is lower than that of the clean EEG signals. This indicates that when the contamination level is high and includes multiple peaks of EOG artefacts, the denoising performance of the ASNet model may be suboptimal.

3.2.2. EMG artefact removal

Table 3 presents the quantitative results of EMG artefact removal performance. The results show that the ASNet model significantly outperforms other competing models in terms of RRMSE, CC, $\text{RRMSE}_{\text{PSD}}$, and MI . The boxplots in Fig. 8 also highlight ASNet's robust performance. In Fig. 9, the performance of EMG artefact removal was quantitatively evaluated at different SNR levels. Although the results differ slightly from those for EOG artefacts, ASNet remains the top-performing model.

As shown in Fig. 10, the best and worst results of removing EMG artefacts using the ASNet model are presented. The figure displays time-domain and frequency-domain segments, particularly highlighting the performance in the 15–120 Hz band. The power spectral density (PSD) of EEG signals contaminated by EMG artefacts is significantly higher than that of clean EEG signals. After denoising, the PSD in the 15–120 Hz band is substantially reduced. The PSD of the denoised EEG signals in this band closely matches that of the clean EEG signals, demonstrating the effectiveness of the ASNet model in suppressing high-frequency artefacts. However, potential failure cases still exist. As shown, in the worst-case scenario, there is a noticeable discrepancy between the waveforms of the denoised EEG signals and the clean EEG signals in the time domain. In the frequency domain, the PSD of the denoised EEG signals in the 15–120 Hz band is lower than that of the clean EEG signals. This indicates that when the entire segment is severely contaminated by high-frequency EMG artefacts, the denoising performance of the ASNet model may degrade.

3.2.3. Mixed artefacts removal

Table 4 presents the quantitative results of removing mixed artefacts (including ocular artefacts EOG and electromyographic artefacts EMG). The comprehensive analysis shows that the ASNet model significantly outperforms other competing models in metrics such as RRMSE, CC, $\text{RRMSE}_{\text{PSD}}$, and MI . The box plots in Fig. 11 further validate ASNet's robust performance in handling mixed artefacts. Additionally, Fig. 12

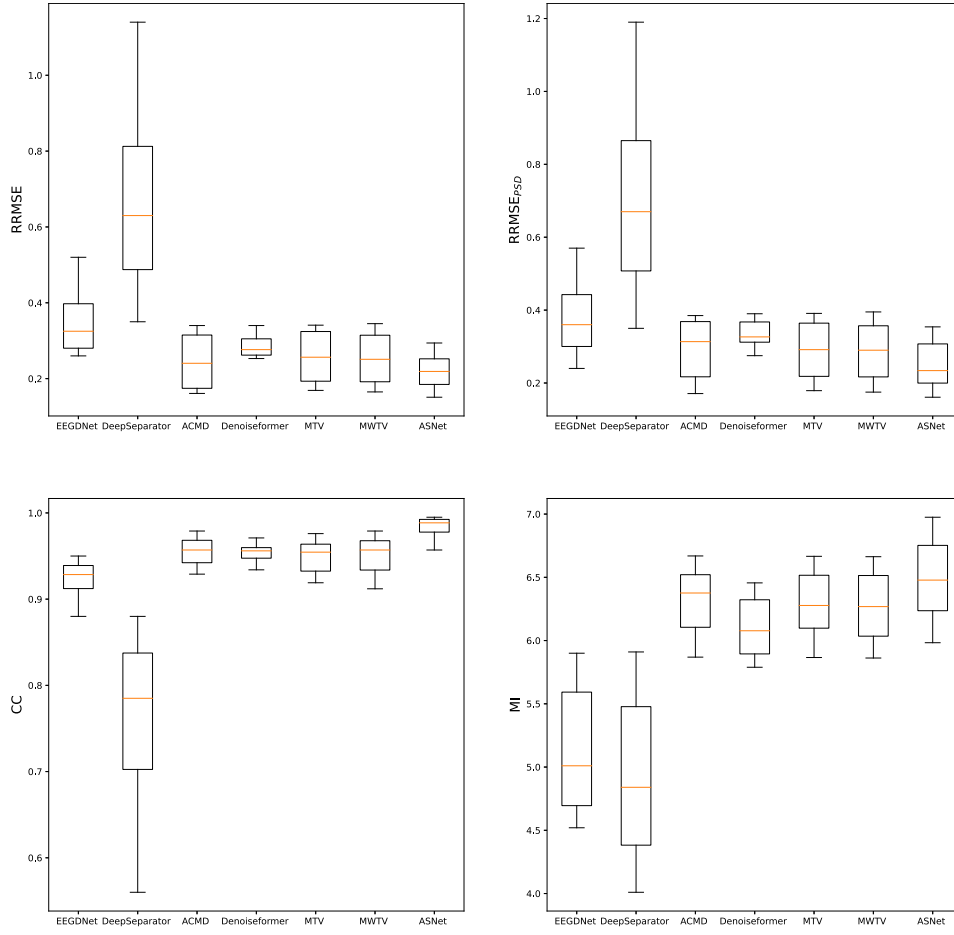


Fig. 5. EOG artefact removal performance. Boxplots compare ASNet's performance with other methods using metrics like RRMSE, CC, RRMSE_{PSD}, and MI. ASNet shows more concentrated distributions, indicating stronger robustness in EOG artefact removal.

quantitatively evaluates the performance of mixed artefact removal under different signal-to-noise ratio (SNR) levels. The results indicate that although handling mixed EOG and EMG artefacts is more complex than dealing with either type of artefact alone, ASNet still maintains optimal performance. As shown in Fig. 13, the best and worst results of using the ASNet model to remove mixed EOG and EMG artefacts are presented. The figure displays time-domain and frequency-domain segments, particularly focusing on the performance in the 0–10 Hz and 15–120 Hz bands. The power spectral density (PSD) of EEG signals contaminated by artefacts is significantly higher than that of clean EEG, but after denoising, the PSD in both the 0–10 Hz and 15–120 Hz bands decreases significantly. Within the 0–80 Hz band, the PSD of the denoised EEG is consistent with that of the clean EEG, demonstrating the effectiveness of the ASNet model in mixed artefact removal. However, potential failure cases remain unavoidable. As shown in the figure, in the worst-case scenario, there are discrepancies between the waveform of the denoised EEG and the clean EEG in the time domain; in the frequency domain, the PSD of the denoised EEG within the 0–80 Hz band is lower than that of the clean EEG. This suggests that when EEG segments are severely contaminated by mixed artefacts, the denoising performance of the ASNet model may be suboptimal.

3.3. Real dataset results

This experiment was conducted using the BCI Competition 2008 Graz Dataset A for training and testing a four-class motor imagery classification task. Ang et al. [52] proposed a novel filter bank common spatial pattern (FBCSP) method capable of automatically selecting key spatiotemporal discriminative EEG features, while Sherwani et al. [53]

Table 4

Comparison of Mixed removal results. The mean $\pm$ std of four metrics from test samples are presented.

Metric	RRMSE	RRMSE _{PSD}	CC	MI
EEGNet	0.437 $\pm$ 0.041	0.498 $\pm$ 0.042	0.884 $\pm$ 0.059	4.897 $\pm$ 0.678
Denoiseformer	0.319 $\pm$ 0.018	0.391 $\pm$ 0.019	0.930 $\pm$ 0.029	5.824 $\pm$ 0.251
DeepSeparator	0.669 $\pm$ 0.077	0.723 $\pm$ 0.077	0.691 $\pm$ 0.156	4.181 $\pm$ 0.941
ACMD	0.294 $\pm$ 0.021	0.332 $\pm$ 0.020	0.928 $\pm$ 0.015	6.083 $\pm$ 0.325
MTV	0.307 $\pm$ 0.020	0.359 $\pm$ 0.022	0.925 $\pm$ 0.015	5.988 $\pm$ 0.306
MWTV	0.310 $\pm$ 0.017	0.352 $\pm$ 0.018	0.922 $\pm$ 0.013	6.003 $\pm$ 0.287
ASNet	0.278 $\pm$ 0.016	0.297 $\pm$ 0.016	0.949 $\pm$ 0.012	6.168 $\pm$ 0.321

combined wavelet transform analysis with neural networks (NN) for EEG classification. In this study, we applied the ASNet model to denoise the raw data and used the undenoised data as a control. Specifically, motor imagery EEG features were extracted using the FBCSP method and wavelet transform, followed by classification in the Keras framework. As shown in Table 5, after applying the ASNet model for denoising, the classification performance improved significantly: Accuracy, Kappa, Precision, and Recall increased by 0.006, 0.008, 0.013, and 0.005, respectively. These results indicate that the denoising effect of ASNet effectively enhances the performance of real datasets in subsequent classification tasks.

To address the performance gap between real and semi-synthetic datasets, we fine-tuned the ASNet model using the motor imagery dataset. Specifically, the original EEG was set as the target output, and the noisy EEG was used as input to continue training the pre-trained ASNet model. As shown in Table 6, the fine-tuned model demonstrated

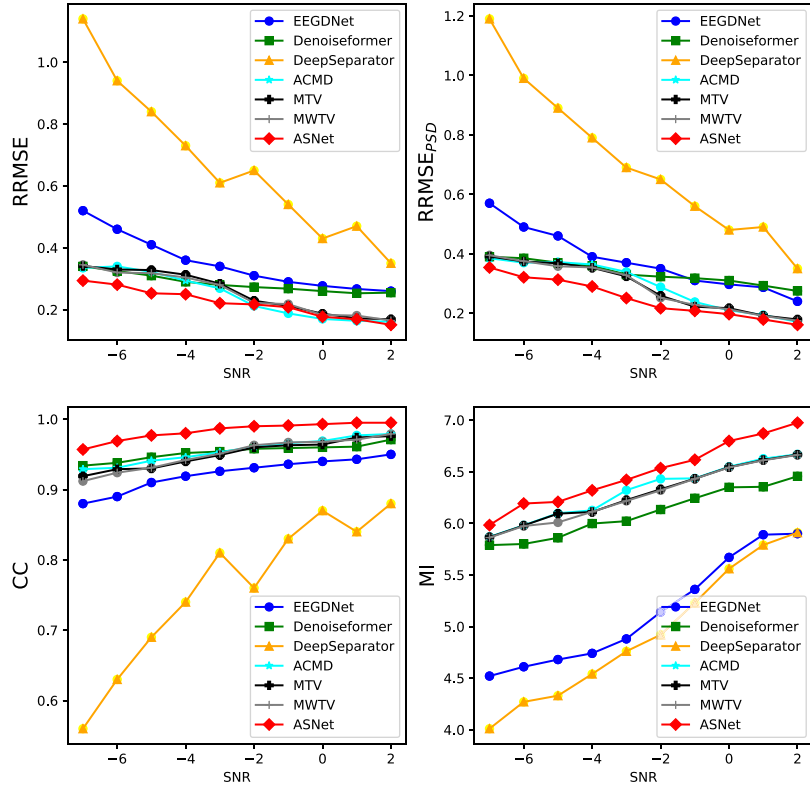


Fig. 6. Denoising robustness across SNR Levels for EOG artefacts. The model's performance is evaluated at multiple SNR levels (−7 dB to 2 dB). ASNet consistently outperforms other methods, maintaining lower RMSE and $RMSE_{PSD}$, and higher CC and MI, even under low SNR conditions.

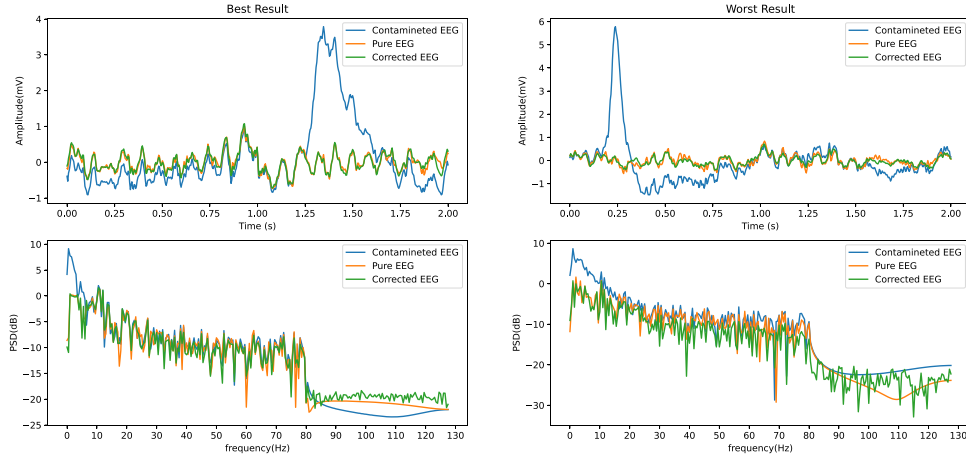


Fig. 7. The figure shows EOG artefacts removal using ASNet, focusing on the 0–10 Hz and 0–80 Hz bands. Denoised signals closely match clean ones after processing, but worst-case scenarios show discrepancies, indicating limitations under high contamination.

significantly better performance in the four-class motor imagery classification task compared to the pre-fine-tuned model: Accuracy, Kappa, Precision, and Recall increased by 0.015, 0.020, 0.016, and 0.029, respectively. Furthermore, we displayed sample segments before and after fine-tuning in both the time and frequency domains (see Fig. 14). The results show that the EEG signals denoised by the fine-tuned model exhibited a significant reduction in PSD in the 0–8 Hz band, while the PSD in the 8–30 Hz band increased. This indicates that the fine-tuned ASNet model not only effectively removed artefacts but also better preserved the true EEG signals. In summary, the fine-tuned ASNet model significantly enhanced denoising performance and classification accuracy on real datasets.

Table 5

Results of the motor imagery experiment. The mean $\pm$ std of four metrics from test samples are presented.

	Accuracy	Kappa	Precision	Recall
Before denoising	0.630 $\pm$ 0.013	0.504 $\pm$ 0.019	0.645 $\pm$ 0.023	0.631 $\pm$ 0.008
After denoising	0.636 $\pm$ 0.011	0.512 $\pm$ 0.015	0.658 $\pm$ 0.017	0.636 $\pm$ 0.019

3.4. Ablation experiment

To comprehensively evaluate the importance of the three components in the ASNet model — Gated Linear Unit (GLU), Channel and Spatial Attention (Attention), and Separator — we conducted ablation

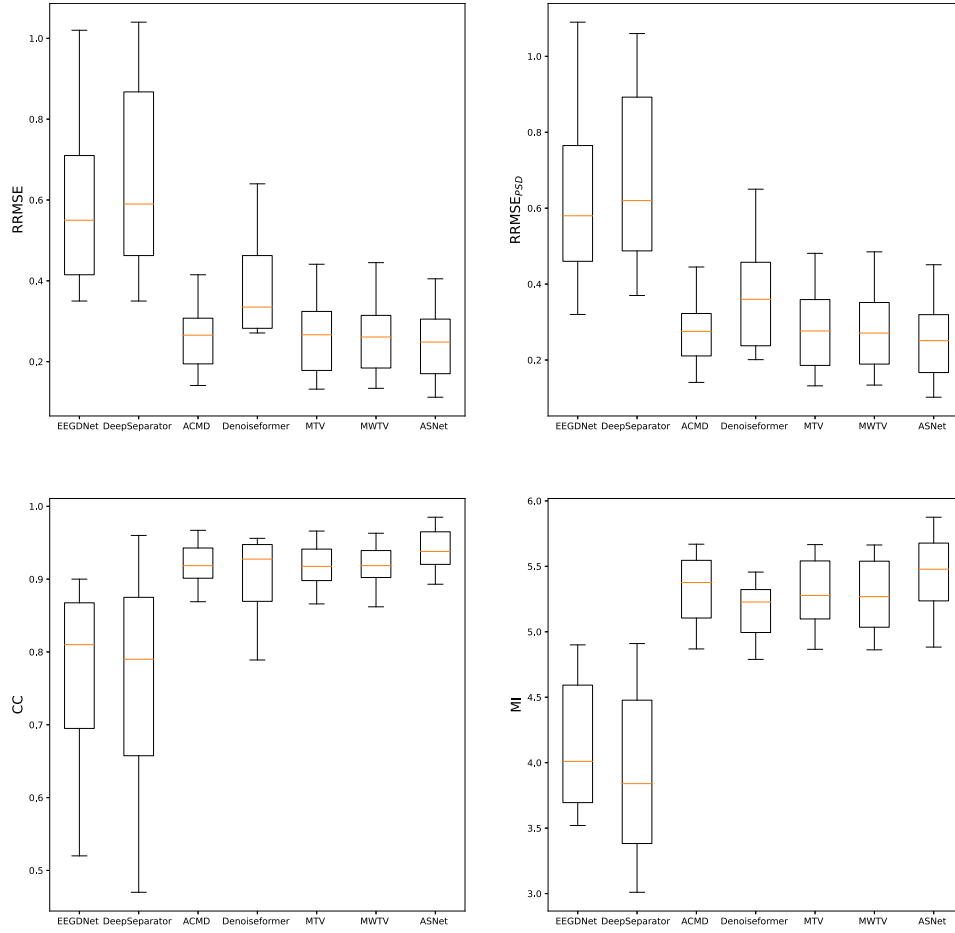


Fig. 8. Boxplots of EMG artefact removal performance. The boxplots compare ASNet's performance with other methods using metrics such as RRMSE, CC, RRMSE_{PSD}, and MI. ASNet shows more concentrated distributions, highlighting its robustness in EMG artefact removal.

Table 6

Fine-tuned ASNet model performance on motor imagery classification. The mean $\pm$ std of four metrics from test samples are presented.

	Accuracy	Kappa	Precision	Recall
Denosed before fine-tuning	0.636 $\pm$ 0.011	0.512 $\pm$ 0.015	0.658 $\pm$ 0.017	0.636 $\pm$ 0.019
Denosed after fine-tuning	0.651 $\pm$ 0.013	0.532 $\pm$ 0.021	0.674 $\pm$ 0.007	0.665 $\pm$ 0.016

Table 7

Results of ablation experiments. The mean $\pm$ std from test samples are presented.

GLU	Attention	Separator	EOG	EMG	p -value(EOG)	p -value(EMG)
✓	✓	✓	0.966 $\pm$ 0.014	0.915 $\pm$ 0.021		
✓	✓	✓	0.956 $\pm$ 0.021	0.881 $\pm$ 0.035	0.0461	0.0121
✓	✓	✓	0.943 $\pm$ 0.023	0.871 $\pm$ 0.036	0.0324	0.0061
✓	✓	✓	0.927 $\pm$ 0.018	0.843 $\pm$ 0.032	0.0146	0.0034

experiments and analysed the statistical significance of performance changes before and after removing each component using t-tests. As shown in Table 7, removing GLU had a relatively small impact, with correlation coefficients for EOG artefacts and EMG artefacts decreasing by 0.01 and 0.034 (p-values of 0.0461 and 0.0121, respectively). Removing Channel and Spatial Attention had a greater impact, with correlation coefficients decreasing by 0.023 and 0.044 (p-values of 0.0324 and 0.0061). Removing the Separator module had the most significant impact, with correlation coefficients decreasing by 0.039 and 0.072 (p-values of 0.0146 and 0.0034). These results indicate that the proposed components are crucial for improving the ASNet model's ability to remove EEG artefacts, and their contributions are statistically significant.

3.5. Model parameters and inference time

The sub-modules in the encoder and decoder (CABS), which contain multiple convolutional attention modules, and the separator use a combined module (TCBS) with multiple temporal convolutional modules. The number of modules contained in the CABS and TCBS have an impact on the model effect. With repeated stacking of small modules in the module, the model is boostable, but with an upper limit. As shown in Table 8, the correlation coefficients in the EOG and EMG removal metrics of the model are boosted by 0.04 and 0.07 respectively for the number of modules from 2 to 4. The number of modules is boosted by 0.08 and 0.07 respectively from 4 to 8, and the number of modules decreases by 0.06 and 0.03 respectively from 8 to 16, which

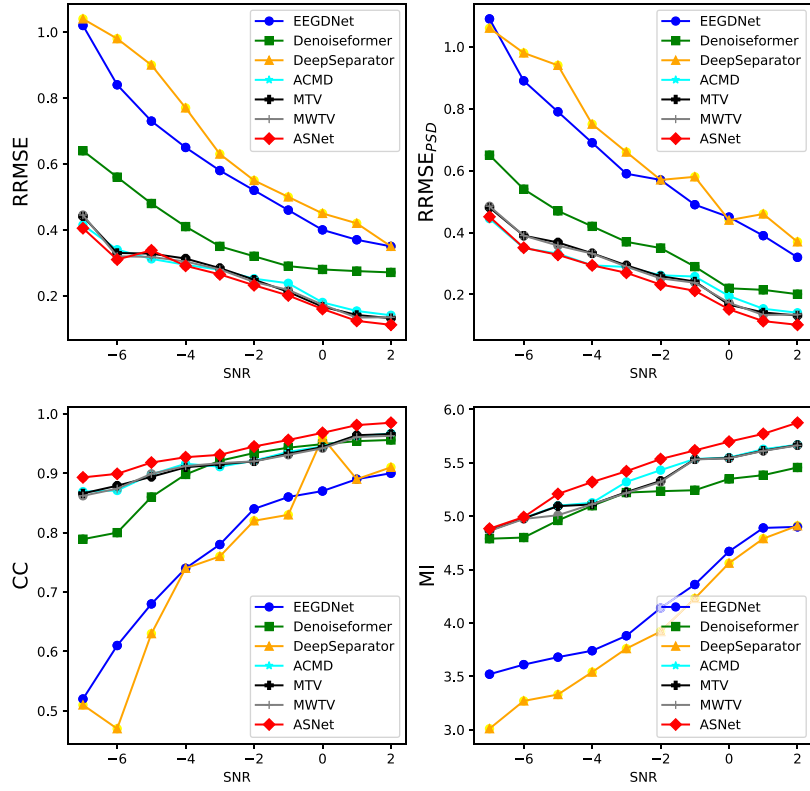


Fig. 9. Denoising robustness across SNR levels for EMG artefacts. The performance of EMG artefact removal is evaluated at different SNR levels (−7 dB to 2 dB). Although EMG artefacts are more complex than EOG artefacts, ASNet consistently outperforms other methods.

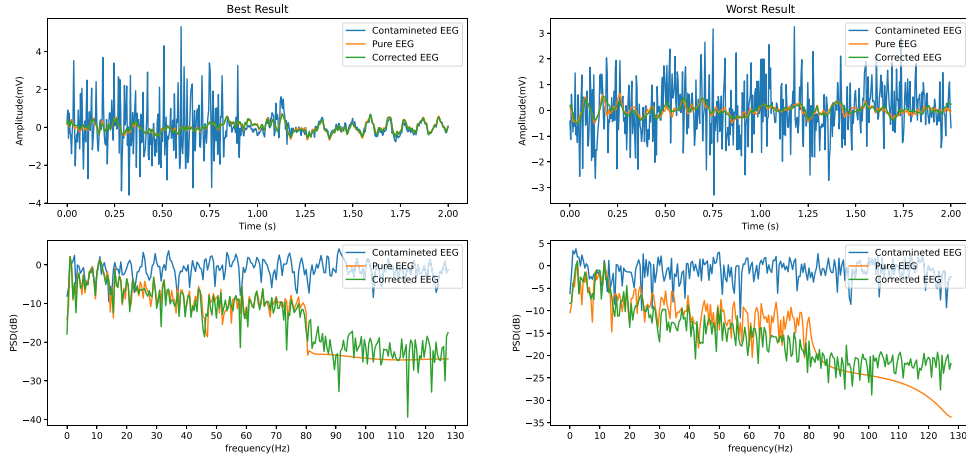


Fig. 10. The figure presents EMG artefacts removal using ASNet, focusing on the 15–120 Hz band. After denoising, PSD is reduced and approaches clean levels, but severe contamination can still cause discrepancies.

shows that the model achieves the best performance for EOG and EMG removal when the number of modules is 8. To measure the goodness of the EEG denoising model, in addition to the two metrics (relative root mean square error and correlation coefficient), the number of parameters and inference time of the model are equally important, especially for portable EEG devices, which require not only effective denoising but also real-time performance. As shown in Table 9, the ASNet model reduces the number of parameters by 7.88 mb and the inference time by 22.2 ms compared to the Denoiseformer model which is close to the same effect, which has lower number of parameters and less inference time, and the improvement of the Transformer model reduces the number of parameters and the computation amount.

Table 8

Parameter experiment results. The mean $\pm$ std from test samples are presented.

Number of modules	EOG	EMG
8	0.966 $\pm$ 0.011	0.915 $\pm$ 0.016
2	0.954 $\pm$ 0.014	0.901 $\pm$ 0.019
4	0.958 $\pm$ 0.022	0.908 $\pm$ 0.021
16	0.960 $\pm$ 0.026	0.912 $\pm$ 0.017

4. Discussion

In this study, we proposed an Adaptive Separation Network (ASNet) based on convolutional attention mechanisms for artefact removal in

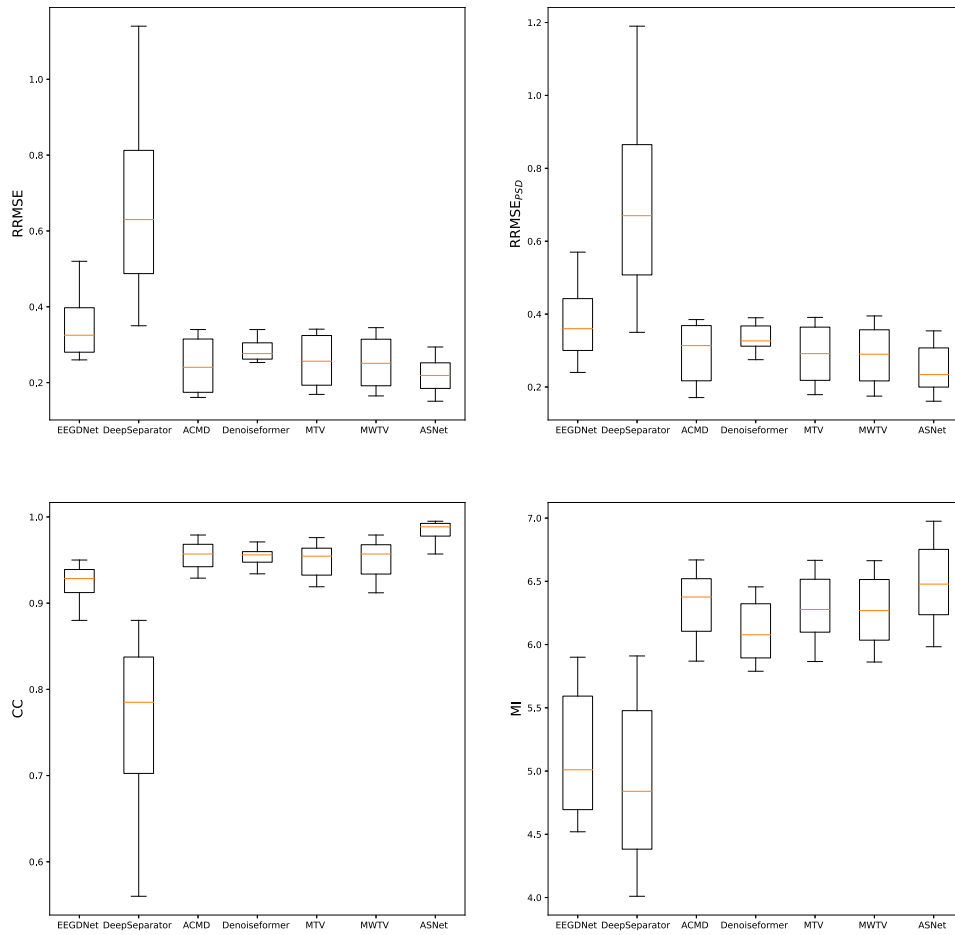


Fig. 11. Boxplots of mixed artefacts removal performance. The box plots further validate ASNet's robust performance in handling mixed artefacts. Compared to other methods, ASNet shows more consistent and superior results across all evaluation metrics.

Table 9

Comparison of number of parameters and reasoning time.

Method	Parameter(million)	Inference time(ms)
EEGDNet	0.18	1.97
Denoiseformer	8.06	24.17
DeepSeparator	0.68	2.75
ASNet	5.48	4

EEG signals. The method adopts an improved U-net architecture by designing a separator module containing multiple dilated convolutions to constrain latent features, effectively separating clean EEG signals from noise. This approach can adaptively capture discriminative features while overcoming the high computational complexity of traditional Transformer architectures, making it suitable for deployment on mobile devices or in resource-constrained environments.

Specifically, ASNet introduces a novel convolutional attention module that combines depthwise separable convolutions with various attention mechanisms. This reduces the number of parameters and computational load while enhancing the ability to learn both local and global features. The module is integrated into both the encoder and decoder, allowing the model to better capture temporal dependencies in EEG signals. Experimental results show that ASNet demonstrates significant artefact removal performance on both semi-synthetic and real datasets, with low computational complexity, providing an efficient method for automatic and real-time preprocessing of EEG signals.

Most signals in the real world are non-stationary, meaning their characteristics change over time. To address this issue, Pachori proposed time-frequency analysis (TFA) [54] techniques such as wavelet

transforms and empirical mode decomposition in his book “Time-Frequency Analysis Techniques and Applications”. For EEG signals, this non-stationarity is particularly pronounced due to the highly dynamic nature of brain activity. Therefore, future research could integrate TFA methods into the ASNet model to help it more accurately capture and interpret signal information. For instance, wavelet transforms or other signal decomposition methods can be used to decompose one-dimensional input data before feeding it into the ASNet model.

To bridge the performance gap between real and semi-synthetic datasets, we fine-tuned the ASNet model using a motor imagery dataset. Specifically, we used raw EEG as the output target and noisy EEG as input to continue training the previously obtained ASNet model for fine-tuning. The fine-tuned model showed significant improvement in motor imagery classification tasks, not only enhancing denoising effects but also better preserving genuine EEG signals. Additionally, we presented sample segments before and after fine-tuning in both time and frequency domains. The results demonstrated that the PSD of denoised EEG signals in the 0–8 Hz band was significantly reduced, while the PSD in the 8–30 Hz band increased, indicating better signal fidelity.

However, experiments also revealed some potential issues. In the worst-case scenarios, when EEG signals are severely contaminated, the denoising performance of the ASNet model is suboptimal. For example, in cases with high EOG artefact peaks, the denoised EEG waveforms show significant differences from clean signals in the time domain; under severe EMG artefact contamination, the PSD of denoised EEG in the 15–120 Hz band is lower than that of clean signals. These results suggest that although ASNet performs well in most cases, there is still

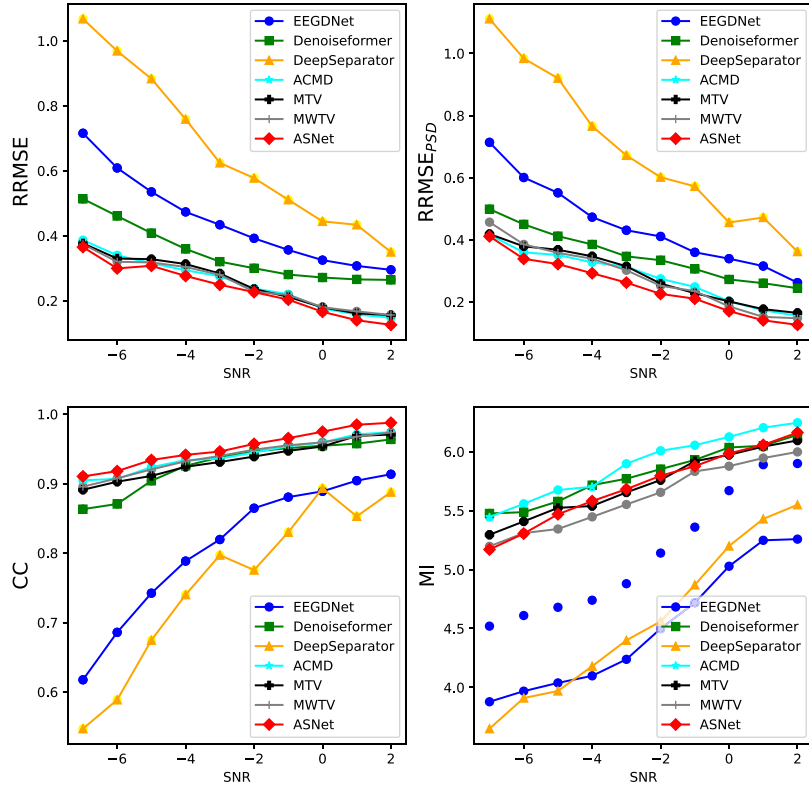


Fig. 12. Denoising robustness across SNR Levels for mixed artefacts. The performance of mixed artefact removal is quantitatively evaluated at different SNR levels. Despite the increased complexity of mixed EOG and EMG artefacts, ASNet maintains optimal performance.

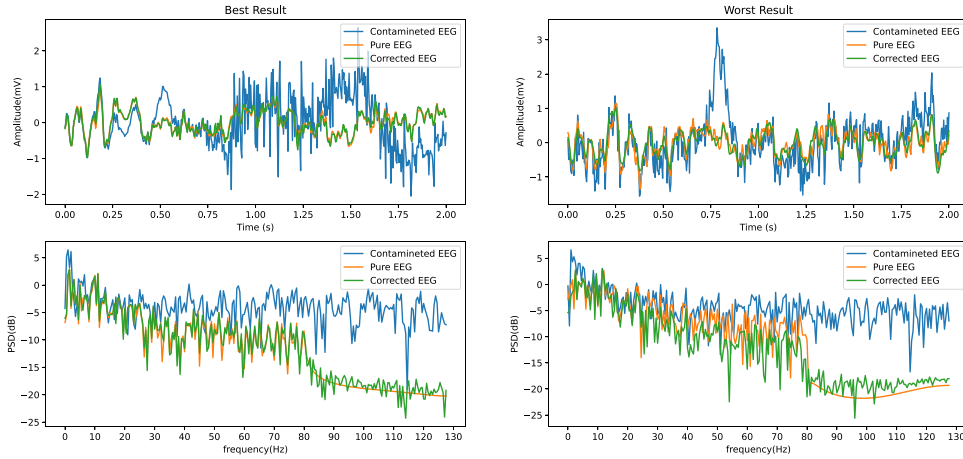


Fig. 13. The figure shows mixed artefacts removal, focusing on 0–10 Hz and 15–120 Hz bands. Well-denoised EEG signals match clean signals, but in the worst case, significant discrepancies occur, especially in the 0–80 Hz band.

room for improvement in handling extreme contamination. Moreover, ASNet does not fully account for the complex spatial information between EEG channels. Designing a model that captures spatiotemporal relationships between channels could further enhance the purity of EEG signals. Therefore, to address various challenges in practical applications, further optimization of the model structure is necessary to improve its robustness and adaptability in extreme contamination scenarios. Through targeted fine-tuning strategies, ASNet can be better adapted to complex real-world scenarios, thereby advancing applications such as EEG-based brain–computer interfaces, ERP analysis, and source localization. This series of studies not only validates the effectiveness of the ASNet model but also provides new directions for future research.

5. Conclusion

In this study, a convolutional attention-based adaptive separation network (ASNet) is designed for EEG artefact removal. Here, based on the U-net architecture, the EEG discriminative features are separated adaptively by adding separators to the latent feature constraints in the latent. In addition, this study improves on the disadvantages of the Transformer itself by designing a novel convolutional attention module to improve the coarse-grained and fine-grained feature interaction learning capability while reducing the number of parameters and computational effort. The experimental results show that ASNet achieves good artefact removal performance on fully synthetic datasets, semi-synthetic datasets, and real datasets, which is verified by two metrics that it has a better removal effect compared to other deep

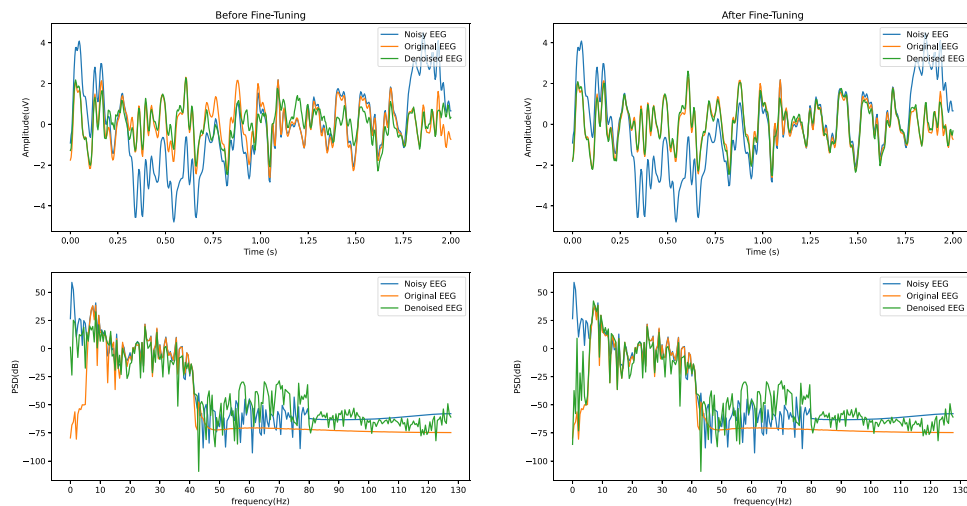


Fig. 14. Comparisons of sample segments before and after fine-tuning are shown in both the time and frequency domains. The results indicate that EEG signals denoised by the fine-tuned model have a significantly reduced PSD in the 0–8 Hz band, while the PSD in the 8–30 Hz band increases. This shows that the fine-tuned ASNet model not only effectively removes artefacts but also better preserves true EEG signals, enhancing the overall quality of the processed EEG data.

learning EEG artefact removal methods. In addition, ASNet has a lower computational complexity compared to other methods, indicating that the method can efficiently extract pure EEG signals from contaminated raw EEG, which is important for automatic and real-time preprocessing of EEG signals.

CRedit authorship contribution statement

Jiaqiang Wang: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Jiaxin Xie:** Writing – original draft, Visualization, Methodology. **Pengrui Li:** Validation, Methodology. **Shihong Liu:** Visualization, Data curation. **Xinmin Ding:** Formal analysis, Data curation. **Jun Chen:** Methodology. **Lutao Wang:** Methodology, Investigation. **Dongrui Gao:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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